Equality and One-Variable Comparison for Positive Collapse Expressions
Normal forms and an effective sign algorithm for the standard Wilken–Weiermann control

28 October 2026
Working manuscript

Abstract

Fix the standard control $\Phi(t) = F_5(t+100)$, with $F_0(t) = 2^t$, in the Wilken–Weiermann collapse below ε_0. Consider positive expressions built from natural constants, positive natural variables, addition and fixed heads $h_\lambda(u) = \psi(\lambda + u)$, where λ is any fixed nonzero limit below ε_0. We give a complete recursive normal form for universal identities in any finite number of variables. In one variable we compute both a finite bound on the exceptional equality set and the eventual sign of every difference. Exact evaluation of the finite prefix then decides all one-variable Boolean combinations of equality and order atoms under a single existential or universal quantifier. The sign algorithm uses exact principal suffix kernels, retained-mass comparisons and a simultaneous lower-profile separation certificate. Its termination is proved by a finite reversed-rank ordinal measure. A source-translation theorem transfers the results to literal canonical assignments of original Gödel-T terms with ground free variables. The identity and one-input results have different scopes; neither supplies multivariate order comparison or arbitrary existential equality elimination.

Contents

1 The problem and the positive language **2**

2 A complete normal form for identities **3**

3 An effective finite-zero bound in one variable **5**

4 Exact suffix kernels and retained mass **6**

5 Paying an entire lower positive profile **7**

6 Adjacent inputs and limit-exponent mass separation **8**

7 Successor cuts and all same-input mass classes **10**

8 The complete one-variable comparison algorithm **11**

9 Consequences for canonical assignments and exact limits **14**

1 The problem and the positive language

Wilken and Weiermann's derivation-length analysis assigns ordinal vectors to terms of Gödel's T [1]. Comparing those particular vectors is a different question from finding some well-founded syntactic ordering of the calculus. The present paper treats a broad scalar fragment of the canonical-assignment comparison problem. The companion ordering paper [3] addresses the syntactic question instead.

Our scalar fragment retains the actual collapse functions. It is not the elementary language generated by $2^f g$: an algorithm or hardness theorem for that elementary language does not automatically transfer to expressions containing $\psi(\lambda + u)$.

Throughout, the norm on Cantor normal forms is additive:

$$\mathrm{N}(0) = 0, \qquad \mathrm{N}(\omega^{\alpha_1} + \cdots + \omega^{\alpha_k}) = \sum_{i=1}^{k} (1 + \mathrm{N}(\alpha_i)). \tag{1.1}$$

Coefficients are written as repeated summands for this definition. Fix

$$F_0(t) = 2^t, \quad F_{j+1}(t) = F_j^{t+1}(t), \qquad f(t) = \Phi(t) = F_5(t + 100).$$

Superscripts on f denote finite iteration, including $f^0(t) = t$.

Definition 1.1 (Positive fixed-limit expressions)**.** For a finite list of variables $x_1, \ldots, x_d$ valued in $\mathbb{N}_{>0}$, the input language $\mathcal{P}_d$ is generated by natural constants, those variables, addition, and $h_\lambda(A) = \psi(\lambda + A)$, for fixed closed nonzero limits $\lambda < \varepsilon_0$. Only finitely many heads occur in any input. Natural coefficient multiplication means repeated addition. There is no subtraction inside a head, multiplication of two open values, variable ordinal index, inverse control or additional orbit-height operation.

A difference $A - B$ is allowed as a signed row to be compared; it is not a new constructor available inside an input expression. Indices may be arbitrarily large below ε_0, and nesting is unrestricted.

Three conclusions will be established. Universal equality is decidable in $\mathcal{P}_d$ for any finite d. In $\mathcal{P}_1$, a nonidentity has only effectively bounded finitely many zeros. Finally, in $\mathcal{P}_1$ its eventual sign is effective. The third conclusion requires substantially more than the second: a nonzero integer-valued sequence can change sign infinitely often without ever being zero.

1.1 The foundational interface

We use the effective maximum predecessor, single-predecessor recurrence, closed evaluation and complete terminal decomposition results of the companion foundations paper [2], proved there from the published collapse definition [1, §2]. Stating the interface also fixes the norm and phase conventions used by every kernel below. For a nonzero limit η and a natural budget b, put

$$M(\eta, b) = \max\{\xi < \eta : \mathrm{N}(\xi) \le b\}.$$

It is computable in finite Cantor normal form and satisfies $\mathrm{N}(M(\eta, b)) = b$. For an infinite $\alpha = \eta + n$ the literal next state of collapse evaluation is

$$\alpha \longmapsto M(\eta, f(\mathrm{N}(\alpha))). \tag{1.2}$$

This is strict ordinal descent. If there are $L \ge 1$ such steps before the first finite state and $t = \mathrm{N}(\alpha)$, then

$$\psi(\alpha) = f^L(t) + L. \tag{1.3}$$

The finite case is $\psi(n) = n$. Both the value and the count are computable by this terminating procedure; no feasible running-time claim is made.

We also use the standard-control estimates and the original controlled strictness property:

$$f(v) \geq 2^v > v, \qquad f(v+1) \geq 2^{f(v)+1}, \tag{1.1}$$

$$\alpha < \beta \;\&\; \mathrm{N}(\alpha) \leq f(\mathrm{N}(\beta)) \implies \psi(\alpha) < \psi(\beta). \tag{1.2}$$

The first line follows from the companion's Elementary control properties lemma. It is for the nonnegative inputs used here; all elementary polynomial estimates below are restricted to inputs at least four. The norm lower bound $\mathrm{N}(\alpha) \leq \psi(\alpha)$ is [1, Proposition 2.3]. These statements do not say that ψ is unrestrictedly increasing in its ordinal argument.

Lemma 1.2 (Sparse peaks)**.** *For an infinite input of norm $t \geq 2$, write $Z = f^L(t) = f(s)$ as in (1.3). Then $0 < L \leq s - 1$, and*

$$Z < \psi(\alpha) = Z + L < 2Z < f(s+1). \tag{1.6}$$

Distinct peaks $f(s)$ and $f(u)$ with $u < s$ satisfy $f(u) \leq f(s-1)$.

Proof. Each norm update increases the norm by at least one. Thus $s = f^{L-1}(t) \geq t + L - 1 \geq L + 1$. Since $f(s) > s$, the count is smaller than Z. The final strict inequality follows from (1.1). The last assertion uses strict increase of the integer map f. □

2 A complete normal form for identities

Definition 2.1 (Collected recursive normal form)**.** A normal form consists of a constant $c \in \mathbb{N}$, a vector $b \in \mathbb{N}^d$, and a finite dictionary of tags (λ, A) with positive natural coefficients. Each A is itself a normal form with nonempty variable support. Its value is

$$c + \sum_i b_i x_i + \sum_{(\lambda,A)} a_{\lambda,A} h_\lambda(A). \tag{2.1}$$

The dictionary graph is finite and acyclic. The height is zero if no tag occurs, and otherwise one plus the largest argument height.

Normalize bottom up. Addition merges dictionaries and coefficient vectors. For a head, normalize its argument; if that argument is closed, evaluate the entire call and return a constant. Otherwise keep the tag $(\lambda, \mathrm{NF}(A))$. Indices are compared as actual canonical Cantor forms. A fixed finite tail in a presented index is first moved into its argument; a zero limit part is removed using $\psi(n) = n$.

Identical normal forms have equal values, by induction. Completeness is a separation theorem and needs more than commutative algebra.

Lemma 2.2 (Orbit cancellation at roots)**.** *Call a natural number an f-root if it is outside* range(f)*. If infinite inputs α, β have root norms t, u and equal peaks, then $\alpha = \beta$.*

Proof. From $f^L(t) = f^J(u)$ cancel the smaller number of iterates, using injectivity. If $L \neq J$, one of the two roots is a positive iterate of the other, a contradiction. Hence $L = J$ and $t = u$. Formula (1.3) gives equal collapse values. If the ordinals differed, order them; their common norm satisfies $t \leq f(t)$, so (1.2) would give distinct values. □

This proof is useful because it avoids any assertion that terminal-peak maps are monotone in arbitrary ordinal inputs.

Theorem 2.3 (Identity normal form)**.** *For $E, F \in \mathcal{P}_d$, the identity $E = F$ holds at every positive valuation exactly when $\mathrm{NF}(E) = \mathrm{NF}(F)$. If the forms differ, one can effectively specify a positive countervaluation of the form*

$$x_i = w_i(f(k) + 1), \qquad w_i, k \in \mathbb{N}_{>0}. \tag{2.2}$$

Consequently universal equality in $\mathcal{P}_d$ is decidable.

Proof. Let $\mathcal{S}$ contain the two root normal forms, zero, and all recursively reachable tag arguments. Let M be the largest raw-variable coefficient in this finite set, $B = \max(2, M + 1)$, and $w_i = B^{i-1}$. Uniqueness of finite base-B notation shows that different coefficient vectors in $\mathcal{S}$ have different weighted slopes $b \cdot w$. On the ray $x_i = w_i n$, any open positive expression is at least n: this follows by induction from positivity and $h_\lambda(q) \geq q$.

We first make every reachable open atom's initial norm a root on the ray $n = f(k) + 1$. For a tag (λ, A) write that norm as

$$t = A_0 + Dn + \sum_j a_j h_{\lambda_j}(A_j), \quad A_0 = \mathrm{N}(\lambda) + c_A \geq 2, \quad D = b_A \cdot w, \quad C = \sum_j a_j. \tag{2.3}$$

If $C = 0$, openness forces $D \geq 1$. For $k \geq \max(4, A_0 + 2D)$,

$$f(k) < t \leq (A_0 + 2D) f(k) < f(k + 1),$$

by (1.1). If $C > 0$, let $Y = f(s)$ be the largest child peak. Every child is open, so $Y \geq f(n) \geq 2^n$. For $n \geq \max(4, A_0, D + 1, 2C + 1)$ the affine part is at most $n^2 \leq Y$. The largest child contributes more than Y, while every child is below $2Y$ by Lemma 1.2. Therefore

$$Y < t < (2C + 1)Y < f(s + 1). \tag{2.4}$$

This too makes t a root. All these bounds are finite syntax data.

Induct on the height of forms in $\mathcal{S}$. Distinct affine forms are unequal for all sufficiently large n, because their slopes differ or their constants differ. Suppose distinct lower-height forms have already been separated. Two distinct tags at the next height cannot have equal peaks: Lemma 2.2 would identify the ordinals $\lambda + A(x)$ and $\mu + B(x)$, hence their limit parts and finite tails. The indices would agree and the different lower-height arguments would have equal values, contradicting induction.

Now cancel identical tags in the difference of two distinct forms at this height. If tags remain, write the row on the weighted ray as

$$\sum_j c_j h_{\lambda_j}(A_j) + pn + q, \tag{2.5}$$

where $c_j \neq 0$ and the remaining tags are distinct. Let $Y = f(s)$ be the largest peak. Its coefficient has absolute value at least one. All other peaks are at most $U = f(s - 1)$, and their full values are below $2U$. Because the largest tag is open, $s \geq n$, while $U \geq s$. Set $K = 2\sum_j |c_j| + |p| + |q|$. The absolute value of all lower contributions is at most KU. Once $k \geq \max(4, K)$, the leading peak satisfies $Y \geq 2^{U+1} > KU$. Thus the row is nonzero, whichever tag is largest. No algorithm for determining that largest tag has been assumed.

For explicit effectivity take the maximum of the root bounds above and, for every pair in $\mathcal{S}$, the displayed row bound K; for an affine row $pn + q$ with $p \neq 0$, also include $|q| + 1$. This single computable k suffices at every induction height. If there are no open variables, evaluate the two constants instead. Normalization and all finite maxima terminate, proving the theorem. □

Example 2.4 (The exact boundary of positivity)**.** The collected forms of $h_\omega(x+y)+x$ and $x+h_\omega(y+x)$ agree. In contrast the two tags in $h_\omega(x)+h_\omega(y)$ remain distinct; the theorem decides an identity, not whether they happen to coincide at some valuation. Inner subtraction cannot be added silently. The literal finite-ω evaluation gives the exact identity

$$h_{2\omega}(n) = h_\omega(h_\omega(n+2)-3)+1 = f(f(n+4))+2.$$

The right side's subtraction is forbidden by Definition 1.1. The free normal form is not asserted for that larger language.

3 An effective finite-zero bound in one variable

From now on every open input depends on the same $x \in \mathbb{N}_{>0}$. This hypothesis permits a stronger separation statement along the entire tail, without selecting the sparse ray (2.2).

Lemma 3.1 (Common-band or root separation)**.** *For any finite collection of normal forms in $\mathcal{P}_1$, a computable N has the following properties for every $x \geq N$. Every affine open atom input lies in $[x, f(x))$. Every nonlinear atom input is a root strictly greater than $f(x)$. Equal peaks of two tags force equal indices and equal argument values.*

Proof. Write an atom norm as $A_0 + bx + \sum a_j h_{\lambda_j}(A_j)$ as in (2.3). For an affine input, $b \geq 1$ and $x \leq A_0 + bx < f(x)$ once $x \geq \max(4, A_0 + b + 1)$. For a nonlinear input the largest child peak Y is at least $f(x)$; for $x \geq \max(4, A_0 + b, 2C + 1)$ the argument proving (2.4) applies at every such x. It is a root greater than $f(x)$.

Two distinct inputs in $[x, f(x))$ cannot lie on the same f-orbit, since the first positive iterate of the smaller is already at least $f(x)$. Two nonlinear roots are handled by cancellation. In a mixed pair, orbit cancellation either puts the larger nonlinear root in the range of f, expresses the smaller affine input as an iterate of that larger input, or identifies their disjoint bands; each is impossible. Thus equal peaks force equal initial norms and equal iterate counts. Equation (1.3) and controlled strictness then identify the two ordinals, exactly as in Lemma 2.2. □

Theorem 3.2 (Effective finite zeros)**.** *Given $E, F \in \mathcal{P}_1$, one computes $N \geq 1$ such that either their normal forms agree, or $E(x) \neq F(x)$ for every $x \geq N$. In the latter case their equality set is the computable finite set obtained by exact evaluation at $1, \ldots, N-1$.*

Proof. Use the same finite set $\mathcal{S}$ as in Theorem 2.3. Take all bounds from Lemma 3.1. For each pair in $\mathcal{S}$, cancel its tag dictionaries. If a tag remains, include 4 and $2\sum|c_j| + |p| + |q|$ in the maximum; otherwise include $|q| + 1$ when its affine slope p is nonzero. Height induction now separates distinct argument forms and, by Lemma 3.1, the peaks of distinct tags. The sparse leading-peak estimate (2.5) separates the full rows on this entire tail. These finitely many bounds are computed before the induction and work simultaneously at every height. □

Lemma 3.3 (Range cells and signed rows)**.** *The following assertions have effective tail bounds. For a nonlinear positive expression E, let Z_i be its displayed open tag peaks, and $Y = \max_i Z_i = f(s)$. Then*

$$Y < E < f(s+1), \qquad E < h_\omega(Y). \tag{3.1}$$

For a value $V \in \mathrm{range}(f)$ on that tail,

$$V < E \iff (\exists i\ Z_i \geq V), \qquad V > E \iff (\forall i\ Z_i < V). \tag{3.2}$$

After cancellation of identical tags in a signed row, its eventual pointwise sign is the coefficient sign of its greatest remaining peak, once those peaks have been ordered. If no tags remain, the row is affine.

Proof. Every open child uses the same input x, so its peak dominates the entire fixed affine part on an effective tail. The proof of (2.4), with the outer index norm omitted, gives $Y < E < (2C+1)Y < f(s+1)$. Increasing the threshold makes this fixed multiple smaller than $h_\omega(Y) = f(Y+2)+1$. There is no range value strictly between $f(s)$ and $f(s+1)$, proving (3.2), including impossibility of $V = E$. Theorem 3.2's height induction separates the remaining tag peaks, and the estimate in (2.5) proves the signed-row assertion. It does not identify the greatest peak; that task is supplied below. □

4 Exact suffix kernels and retained mass

The sign algorithm uses auxiliary kernels to expose the terminal peak of a fixed source head. Their indices and masses are closed finite data, and auxiliary kernels are not new input constructors in $\mathcal{P}_1$.

Definition 4.1 (Principal suffix kernel)**.** Let $\pi = \omega^\beta \geq \omega$ be a fixed principal, $a \in \mathbb{N}$, and $t \geq a + \mathrm{N}(\pi)$. Start at suffix $\pi + (t - a - \mathrm{N}(\pi))$, with total norm t and permanent numerical mass a. At a nonfinite suffix $\eta + n$, replace it by $M(\eta, f(v) - a)$, where v is the current total norm. Let $E_{\pi,a}(t)$ count steps to the first finite suffix and put

$$K_{\pi,a}(t) = f^{E_{\pi,a}(t)}(t). \tag{4.1}$$

The finite exit suffix is $K_{\pi,a}(t) - a$. The value in (4.1) is the *total* exit norm, and always belongs to range(f).

The suffix is strictly descending and its next norm budget is always genuine. Thus the procedure terminates. A numerical mass need not be the norm of any source prefix.

Lemma 4.2 (Exact prefix splitting)**.** *Let κ be a fixed Cantor prefix and let η be a nonzero limit whose leading monomial is at most the least monomial of κ; the zero prefix is also allowed. In particular one may take $\kappa = \pi k$ and* $\mathrm{lead}(\eta) \leq \pi$*. For $b \geq c = \mathrm{N}(\kappa)$,*

$$M(\kappa + \eta, b) = \kappa + M(\eta, b - c). \tag{4.2}$$

A descent above κ reaches a state $\kappa + n$ with finite remainder before passing below κ. During that segment the remainder's exact norm control is

$$G_c(v) = f(c + v) - c \geq f(v) > v. \tag{4.3}$$

Proof. The ordinal κ is eligible below $\kappa + \eta$, so the maximum cannot have a smaller prefix. Every ordinal in the interval has a unique remainder $r < \eta$, and compatible Cantor concatenation gives norm $c + \mathrm{N}(r)$. Maximizing that remainder proves (4.2). Equal leading boundary exponents cause coefficients and their norm contributions to add, so they are allowed. Budgets increase, keeping the prefix affordable. Equation (4.2) forbids skipping the finite-remainder cut; ordinal well-foundedness supplies it. Norm saturation proves (4.3). An increasing integer map preserves integer differences, giving its inequality. □

Lemma 4.3 (Ordered controls and counts)**.** *For increasing integer controls $g \leq h$, both strictly exceeding their input, let L_g, L_h be the respective norm-maximal descent counts. Then*

$$\alpha \leq \beta, quad \mathrm{N}(\alpha) \leq \mathrm{N}(\beta) \quad \Longrightarrow \quad L_g(\alpha) \leq L_h(\beta). \tag{4.4}$$

Proof. Induct on β. Finite α has zero count. Otherwise write the limit parts of α, β as λ, μ. If $\lambda = \mu$, their next states p, q are ordered by monotonicity in the budget, and $\mathrm{N}(p) = g(\mathrm{N}\,\alpha) \leq h(\mathrm{N}\,\beta) = \mathrm{N}(q)$. If $\lambda < \mu$, then $\alpha < \mu$ and its norm is less than the next h-budget. Hence α is eligible below μ, so $p < \alpha \leq q$; the same norm inequality holds. Apply induction at $q < \beta$ and add the first step. □

This also proves fixed-head monotonicity: increasing the finite tail of $\rho + n$ increases its norm and does not decrease its count, so its full ψ value is strictly increasing. Fixed-mass kernels strictly increase their genuine input by the same argument. Decreasing their retained mass at fixed total input weakly increases the suffix and its budgets, so cannot decrease the exit norm.

Lemma 4.4 (Exact peak word)**.** *For each fixed limit λ, one computes a finite word W_λ of principal kernels such that the peak of $h_\lambda(A)$ is exactly*

$$Z_{\lambda,A} = W_\lambda(\mathrm{N}(\lambda) + A). \tag{4.5}$$

The omitted count is positive: $h_\lambda(A) = Z_{\lambda,A} + D_{\lambda,A}$.

Proof. Expand λ into coefficient-one principal Cantor blocks. Delete them from right to left. For each block π, the remaining earlier prefix has a fixed norm a. Lemma 4.2 says that this segment changes the current total norm by exactly $K_{\pi,a}$. The last segment has prefix zero. Concatenating the finitely many segments gives the exact word. Every removed nonfinite segment contributes its literal count to D; no count is set to zero in (4.5). □

5 Paying an entire lower positive profile

A lower profile below a principal $L > \omega$ means a fixed expression $P(z)$ formed from constants, z, addition and heads h_ρ with $\rho < L$. Its heads may themselves be nonprincipal limits. The following reserve argument is needed to compare unequal kernel inputs and to handle a limit exponent. Merely knowing that a larger kernel exceeds a fixed number of f-iterates would be insufficient.

Lemma 5.1 (One-stage reserve consumer)**.** *Let $\rho < \pi m$ be a fixed nonzero limit, $m \geq 1$, and begin an ordinary descent at $A = \pi(k+m)+n$, of norm $X \geq x \geq \mathrm{N}(\rho)$. At its first cut $\pi k + n'$ with finite remainder,*

$$n' > J_\rho(x) := f^{D_\rho(x)}(x) + D_\rho(x), \tag{5.1}$$

where $D_\rho(x)$ is the count from $\rho + (x - \mathrm{N}(\rho))$.

Proof. Put $c = k\,\mathrm{N}(\pi)$ and $\alpha = \rho + (x - \mathrm{N}\,\rho)$. The ordinal $\pi k + \alpha$ is below the initial limit part and has norm $c + x \leq 2X < f(X)$, hence is eligible for the first predecessor. Its remainder is at least α and has norm $f(X) - c \geq x$. Lemmas 4.2 and 4.3 show that the following prefix segment takes at least $d = D_\rho(x)$ further steps. Its exit total norm Y is at least $f^{d+1}(X)$. As $c \leq X$ and every increasing integer iterate preserves differences,

$$n' = Y - c \geq f^{d+1}(X) - X \geq f^{d+1}(x) - x.$$

Write $W = f^d(x) \geq x + d$. Then $f(W) - x > 2W - x \geq W + d = J_\rho(x)$, as required. Notice that the prefix mass was paid using $c \leq X$; it need not be bounded by the smaller test input x. □

Lemma 5.2 (Effective profile reserve)**.** *From a lower profile P below $L > \omega$ one computes a nonempty word*

$$F = h_{\rho_l} \circ \cdots \circ h_{\rho_1}, \quad F(z) > P(z) \ (z \geq z_0), \quad R = \sum_j (\mathrm{N}(\rho_j) + 1), \quad H = \max_j \mathrm{N}(\mathrm{lead}(\rho_j)), \tag{5.2}$$

with every $\rho_j < L$. If $\pi \geq \omega$ is principal and at least every displayed leading monomial, a descent from $\pi R + n$, $n \geq z \geq z_0$, has terminal norm greater than $P(z)$. The same conclusion holds with any additional permanent numerical mass c.

Proof. Choose $z_0 \geq 4$, all constants and every weighted addition fan-in in P. Process its expression graph in topological order, maintaining a common upper bound u on all node values already visited. An addition with fan-in b is bounded by $bu \leq u^2 \leq 2^u < h_\omega(u)$; append h_ω to the running bound. For a head node append that head, using its monotonicity. Every update increases the running bound. Append a final h_ω to make the bound strict and the word nonempty. Since $L > \omega$, all added heads are still below L.

For a head ρ_j take $m_j = \mathrm{N}(\rho_j) + 1$ blocks. Its leading coefficient is at most its norm, so $\rho_j < \pi m_j$. Moreover the current norm at $\pi(k + m_j) + n$ is at least $\mathrm{N}(\rho_j) + n$. Lemma 5.1, with $x = \mathrm{N}(\rho_j) + n$, therefore makes the finite tail at the next cut exceed $h_{\rho_j}(n)$. Repeat through the word, using exactly R blocks. The final finite norm exceeds $F(z) > P(z)$. For permanent mass c, the exact suffix control is $G_c \geq f$. Lemma 4.3 orders the counts and the iterated norms, so the terminal suffix norm is at least the unprefixed one; adding c only increases the total norm. □

Lemma 5.3 (Direct kernel growth)**.** *For fixed $L > \omega$ principal and fixed lower profile P, one computes T such that, for $t \geq T$ and every genuine mass $0 \leq c \leq t - \mathrm{N}(L)$,*

$$K_{L,c}(t) > P(t). \tag{5.3}$$

Also, $K_{\pi,c}(t) \leq h_\pi(t)$ on its genuine domain.

Proof. Choose one fixed principal $\omega \leq \pi < L$ dominating the finite list of leading monomials in (5.2). For $t \geq \max(z_0, R\,\mathrm{N}\,\pi, 4, \mathrm{N}\,L)$, the ordinal $S = \pi R + t < L$ has

$$c + \mathrm{N}(S) \leq R\,\mathrm{N}\,\pi + 2t \leq 3t < f(t).$$

It is eligible below the actual first suffix $M(L, f(t) - c)$. Count comparison under the same induced control transfers the reserve bound from S to that maximum. This proves (5.3) uniformly in the masses. For the final assertion first lower the retained mass to zero, then increase the seed from t to $\mathrm{N}\,\pi + t$. Mass monotonicity and fixed-mass input monotonicity bound the kernel peak by the full head value. □

6 Adjacent inputs and limit-exponent mass separation

Theorem 6.1 (Cross-mass strict-input separation)**.** *For a fixed principal $L > \omega$ and fixed lower profile P with reserve data (5.2), put*

$$T = \max(4, R + 5, z_0, \mathrm{N}\,L, H). \tag{6.1}$$

For $t \geq T$, $0 \leq c \leq t + 1 - \mathrm{N}\,L$, and $0 \leq e \leq t - \mathrm{N}\,L$,

$$K_{L,c}(t + 1) > P(K_{L,e}(t) + E_{L,e}(t)). \tag{6.2}$$

There is no ordering requirement on c, e. More generally the same conclusion holds with any $u > t$ and any mass genuine at u on the left.

Proof. Write $B = f(t)$, $B' = f(t+1)$, $b = M(L, B-e)$, $\pi = \mathrm{lead}(b)$ and $w = \mathrm{N}\,\pi$. Since $B - e \geq f(t) - t + \mathrm{N}\,L \geq t + \mathrm{N}\,L \geq H$, every leading monomial needed by (5.2) is eligible below L in this budget. Also $B - e \geq 2$, so $\pi \geq \omega$. Thus $\pi < L$, $w \leq B$, and π dominates every required leading monomial.

Put $p = c + Rw$ and

$$k = B' - B - (p - e), \qquad A = \pi R + b + k.$$

Here $p - e \leq t + 1 + RB \leq (R+1)B$ and $B' \geq 2^{B+1} > (R+2)B$, so $k \geq 0$. The sum is compatible CNF, has norm $B'-c$, and lies below the principal L. Hence $A \leq Q = M(L, B'-c)$ and both have the same norm. The latter is the upper process's actual first suffix.

Follow the packed suffix A retaining its first R blocks while deleting the remainder $b + k$. At the reserve cut it has finite tail

$$n = V - p, \qquad V = f^q(B'). \tag{6.3}$$

The lower process, after its first step, starts at b with norm B and retained mass e. Let its remaining count be ℓ, and set $D = 1 + \ell = E_{L,e}(t)$ and $Z = f^\ell(B) = K_{L,e}(t)$. The packed remainder initially exceeds b. The initial global gap pays the mass difference $p - e$ and, as increasing integer iterates preserve differences, at every simultaneous later step $f^i(B') - p \geq f^i(B) - e$. Monotonicity of M in its limit bound and budget keeps the packed remainder above the lower remainder; it cannot become finite earlier. Hence $q \geq \ell$.

If $s = f^{D-1}(t)$, then

$$V \geq f^\ell(B') = f^D(t+1) \geq f(s+1) \geq 2^{Z+1}.$$

Furthermore $D < Z$ and $p \leq (R+1)Z$. Since $Z \geq t \geq R + 5$, $p + Z + D < (R+3)Z < Z^2 < 2^{Z+1} \leq V$. Thus $n > Z + D$ after the full retained mass has been subtracted. Lemma 5.2 gives terminal total norm greater than $P(Z + D)$ for the packed process. Count comparison from $A \leq Q$ at their equal norms, under their common control G_c, transfers that bound to the actual upper kernel.

For arbitrary $u > t$, use the worst-mass inequalities

$$K_{L,c}(u) \geq K_{L,u-\mathrm{N}\,L}(u) \geq K_{L,t+1-\mathrm{N}\,L}(t+1). \tag{6.4}$$

The first is mass monotonicity. For the second both processes start at the identical suffix L, while their simultaneous suffix budgets are $f^i(u) - u + \mathrm{N}\,L \geq f^i(t+1) - (t+1) + \mathrm{N}\,L$. Their counts and terminal total norms are ordered. Apply the adjacent case with the displayed genuine upper mass. □

At $L = \omega$ the kernel is simply $f(t)$, independently of mass. There are no nonzero lower limit heads, so $P(z) = az + b$. Taking $T_\omega = \max(4, a + b + 2)$ gives $P(f(t) + 1) < f(t+1)$ by (1.1). This is the separate base case of Theorem 6.1.

Theorem 6.2 (Limit-exponent mass separation)**.** *Let $L = \omega^\beta$ with β a nonzero limit. For a fixed lower profile P and data (5.2), put*

$$T = \max(4, R + 4, z_0, \mathrm{N}\,L + 1, H). \tag{6.5}$$

For $t \geq T$ and $0 \leq c < e \leq t - \mathrm{N}\,L$,

$$K_{L,c}(t) > P(K_{L,e}(t) + E_{L,e}(t)). \tag{6.6}$$

Consequently the lexicographically larger pair $(input, -mass)$ gives domination against every fixed lower profile, on an effective genuine tail. Equal pairs give equal values.

Proof. Put $B = f(t)$ and $C = f(B)$. Because β is a limit, the maximum affordable exponent $\delta_a = M(\beta, B - a - 1)$ saturates its norm budget. Its monomial $\pi_a = \omega^{\delta_a}$ already costs $B - a$, so the first suffix below L is exactly π_a, with coefficient one and no remainder. Since $c < e$, saturation gives $\delta_c > \delta_e$ and $\pi_c > \pi_e$. Put $w = \mathrm{N}\,\pi_e = B - e$. The inequality $B - e \geq f(t) - t + \mathrm{N}\,L \geq t + \mathrm{N}\,L \geq H$ shows that π_e is infinite and dominates every required leading monomial.

At the second upper step pack

$$k = C - c - (R+1)w, \qquad A = \pi_e(R+1) + k.$$

Since $C > (R+1)B$ and $c + (R+1)w \leq (R+1)B$, the tail k is nonnegative. The ordinal A is below π_c and has norm $C - c$, hence is below the actual second upper suffix $Q = M(\pi_c, C - c)$ at that same norm.

In the packed process hold R blocks while deleting $\pi_e + k$. The retained mass is $p = c + Rw$. Compare this remainder with the lower process after its first step, namely π_e with total norm B and retained mass e. The global difference $C - B$ exceeds $p - e \leq RB$, so synchronized budgets keep the packed remainder above the lower one. If the latter needs ℓ further steps, the reserve cut has norm $V \geq f^{\ell}(C) = f(Z)$, where $Z = K_{L,e}(t) = f^{\ell}(B)$ and $D = E_{L,e}(t) = 1 + \ell < Z$. Since $p \leq (R+1)Z$ and $Z \geq R + 4$, $f(Z) > (R+3)Z > p + Z + D$. Its finite tail after all retained mass is paid is therefore greater than $Z + D$.

Consume the reserve by Lemma 5.2, still with permanent mass c. Transfer from $A \leq Q$ at equal suffix norms using Lemma 4.3. This yields (6.6). Every estimate uses only the displayed genuine domains, not a fixed bound on c, e, so it is uniform over the stated numerical masses. Combine with Theorem 6.1 for unequal inputs to obtain the lexicographic assertion. A profile-dependent threshold is intended; there is no single threshold valid for all profiles at once. □

The limit-exponent hypothesis is essential. For example, the successor identity proved next gives $K_{\omega^2,0}(t) = f(K_{\omega^2,2}(t))$, while the lower profile $h_\omega(z) = f(z+2) + 1$ exceeds $f(z)$. Thus arbitrary lower-profile absorption at the same input is false for successor exponents.

7 Successor cuts and all same-input mass classes

Lemma 7.1 (Exact successor cut and count cocycle)**.** *Let $\theta = \omega^{\gamma} \geq \omega$, $L = \theta\omega$, and $w = \mathrm{N}\,\theta$. On the common genuine domain $t \geq c + w + \mathrm{N}\,L$,*

$$K_{L,c}(t) = K_{\theta,c}(K_{L,c+w}(t)), \tag{7.1}$$

$$E_{L,c}(t) = E_{L,c+w}(t) + E_{\theta,c}(K_{L,c+w}(t)). \tag{7.2}$$

Proof. For $b = wq + r$, $0 \leq r < w$, the literal maximum is

$$M(L, b) = \theta q + M(\theta, r). \tag{7.3}$$

The largest monomial below L is θ, costing w; greedily maximize its coefficient, then the remainder below θ. This also covers $q = 0$. Lowering the first budget from $f(t) - c$ to $f(t) - c - w$ removes exactly one θ block and leaves the same remainder. Retaining that one block in the c-process makes its mass during deletion of the other remainder exactly $c + w$. Both processes have total norm $f(t)$ after their first step. Prefix splitting therefore gives identical subsequent norms until the latter process's finite cut. The remaining block is exactly $K_{\theta,c}$, and the counts add. The stronger common domain ensures that the inner shifted-mass kernel exists. □

Lemma 7.2 (Small principal kernels)**.** *On their genuine domains,*

$$K_{\omega,c}(t) = f(t), \quad E_{\omega,c}(t) = 1, \qquad E_{\omega^2,c}(t) = f(t)/2 - \lceil c/2 \rceil + 1. \tag{7.4}$$

Thus ω^2 kernels at the same input have exactly the mass classes $\lceil c/2 \rceil$, ordered in reverse. The control has even values.

Proof. Below ω the first predecessor is finite. Below ω^2 it is $\omega q + r$ with $q = \lfloor (f(t) - c)/2 \rfloor$ and $0 \le r < 2$. The first step is followed by exactly one step for each ω block, because each such block's deletion uses a finite suffix maximum. Thus the count is $q + 1$, giving (7.4). The definition of the iterated exponential hierarchy makes each $f(t)$ even. □

Theorem 7.3 (Strict mass order above degree two)**.** *For every fixed principal $L \ge \omega^3$, a computable T_L satisfies*

$$t \ge T_L, quad 0 \le c < e \le t - \mathrm{N}\, L \quad \Longrightarrow \quad K_{L,c}(t) > K_{L,e}(t). \tag{7.5}$$

For a nonzero limit exponent take $T_L = \max(7, \mathrm{N}\, L + 1)$. For $L = \theta\omega$, $\theta \ge \omega^2$, $w = \mathrm{N}\, \theta$, take

$$T_L = \max(4, \mathrm{N}\, L, 2w, \max(8, w), S_\theta), \tag{7.6}$$

where $S_\theta = 0$ at $\theta = \omega^2$ and otherwise $S_\theta = T_\theta$.

Proof. The limit case is Theorem 6.2 with profile $P(z) = z$. Its compiler uses just one final h_ω, giving $R = 3, H = 2, z_0 = 4$. For the successor case let $B = f(t)$, and write the first suffixes from (7.3) as $\theta q_a + r_a$, $a = c, e$. Here $q_c \ge q_e = q$ and $r_a = M(\theta, (B - a) \bmod w)$. The domain and (7.6) imply $B - e \ge f(t) - t + \mathrm{N}\, L \ge 2w$, so $q \ge 2$.

Hold q common θ blocks while deleting residuals $\theta(q_c - q) + r_c$ and r_e. The first residual is at least the second. Their permanent masses are $c + wq$ and $e + wq$; at equal global norms their suffix budgets preserve this order. The finite-cut entry norms therefore satisfy $U_c \ge U_e \ge B$.

With k common blocks remaining, the child masses are $a_k = c + w(k-1) < b_k = e + w(k-1)$. They are genuine even at the smaller entry norm $V \ge B$, because

$$b_k \le e + w(q-1) = B - ((B - e) \bmod w) - w \le B - w \le V - w. \tag{7.7}$$

If the entry norms are unequal, Theorem 6.1 with identity profile preserves strict order, uniformly over both masses; its threshold is $\max(8, w)$. If they agree and $\theta \ge \omega^3$, the inductive strict-mass theorem makes the first child outputs strictly ordered, and every subsequent block preserves it.

For $\theta = \omega^2$, Lemma 7.2 gives weak order, with a tie precisely when $b_k = a_k + 1$ and a_k is odd. If the first block ties, the next pair of masses subtracts $w = 3$ from both; the smaller mass is then even, so their ceilings differ and the second outputs are strictly ordered. There are at least two blocks, and (7.7) pays their domains. All further blocks preserve the strict order. This proves the base and the successor induction. Removing the finite terminal successor tail of the exponent reaches either this base or a limit exponent, so the threshold recursion terminates. □

8 The complete one-variable comparison algorithm

We now supply the order decision that the normal-form proof deliberately did not assume. There are two sorts in the auxiliary comparison language. The source sort P is $\mathcal{P}_1$. A range expression R is a nonempty finite word of kernels applied to one open P expression. Every

argument of a kernel is either P or R; a sum of range expressions is not introduced into a kernel argument. Closed words are evaluated and folded into constants. Every open expression of either sort is at least x on a computable tail. All generated masses and indices are fixed natural or ordinal data on each branch.

Theorem 8.1 (Effective eventual signs)**.** *Given two expressions in these sorts, one computes a sign $\sigma \in \{-1, 0, 1\}$ and $N \geq 1$ such that their difference has sign σ for every $x \geq N$. In particular this holds for every pair in $\mathcal{P}_1$.*

The proof is a simultaneous comparison and separation-certificate algorithm. We specify its rules before proving their common termination measure. Every rule retains the thresholds of all its children and the finitely many kernel domains it uses. Since every open argument is at least x eventually, each fixed genuine-domain obligation is met by increasing N to a computable constant.

8.1 The comparison rules

For P/P, form the signed normal-form row. An affine row is decided directly. If tags remain, replace each relevant tag by its exact peak word (4.5), compare those R expressions recursively and choose the greatest peak. Distinct tags cannot have equal peaks beyond the bound of Theorem 3.2. Return its coefficient sign, using Lemma 3.3. Identical forms return exact equality.

For R/P, a constant or nonconstant affine P is eventually smaller: the open R exceeds $f(x)$, which dominates every fixed affine expression. For a nonlinear P compare R with its displayed peaks (4.5), and use the exact range-cell test (3.2). The P/R rule is symmetric.

For R/R with equal outer principal, compare the two literal arguments recursively. Strictly ordered arguments give the same strict kernel order by Theorem 6.1, or by monotonicity of f at ω. If the arguments agree eventually, use the following same-input key: mass is irrelevant at ω; compare $\lceil c/2 \rceil$ in reverse at ω^2; compare masses in reverse above ω^2 by Theorem 7.3. Equal keys give equal outer values.

For different outer principals $H > M$, if H has a successor exponent, use the exact cut (7.1) on the H side until the new outer principal is M or has a limit exponent still above M. This cannot jump below M: every principal below $\theta\omega$ is at most θ. It takes only the finite terminal successor part of the exponent. Combine these rewrites immediately with the equal-principal rule or the following mixed-limit rule; do not recursively call the unmodified comparison at an intermediate, larger syntax state.

In the mixed-limit case write

$$X = K_{L,c}(A), \qquad Y = K_{M,e}(B), \quad M < L, \tag{8.1}$$

where L has a nonzero limit exponent. Recursively compare X with B. If $X \leq B$, then $Y > B \geq X$. If $X > B$, the separation certificate below, with lower profile h_M, gives $X > h_M(B) \geq Y$, by Lemma 5.3. There is no equality branch in this mixed-limit case.

8.2 The simultaneous limit-cut certificate

Whenever one original side of a comparison is a bare limit-exponent kernel $X = K_{L,c}(A)$ and its result is $X > S$, the algorithm also constructs, for every requested fixed lower profile P below L, a computable modulus $N_{X,S}(P)$ such that

$$X(x) > P(S(x)) \quad (x \geq N_{X,S}(P)). \tag{8.2}$$

This concerns generated P/R expressions S, not arbitrary integers below X: the assertion for $S = X - 1$ would be false, and subtraction does not occur in the generated language.

Here is the certificate's complete case analysis, following the same strictly smaller children as comparison.

1. If $S = ax + b$, use direct growth (5.3) for the fixed profile $z \mapsto P(az + b)$ at the actual argument $A \geq x$. Positivity of P makes this imply (8.2).

2. If S is nonlinear positive, $S < X$ forces every child peak $Z_i < X$ by the range-cell rule. Apply each smaller child's certificate to $P \circ h_\omega$ and use $S < h_\omega(\max_i Z_i)$. Take the finite maximum of their moduli; no permanent choice of maximizing child is needed.

3. If S has the same outer principal L, strict order means either a greater input for X, or equal inputs and smaller mass. The lexicographic conclusion of Theorems 6.1 and 6.2 applies directly to P, at the genuine integer inputs.

4. If $S = K_{M,e}(B)$ has smaller outer principal, its comparison used the smaller child $X > B$. Apply that child's certificate to $P \circ h_M$ and use $S \leq h_M(B)$. This handles any finite nesting of lower contexts by recursive composition, not by an unproved absorption step.

5. If S has larger limit outer principal $H > L$, its comparison with X compares S against A. The only branch giving $X > S$ has $S \leq A$; direct growth for $K_{L,c}$ gives $X > P(A) \geq P(S)$. In the other branch the larger kernel's child certificate would force $S > h_L(A) \geq X$, so cannot yield $X > S$.

6. A larger successor outer principal on S is unfolded exactly until it reaches L or a larger limit exponent; use the corresponding preceding case. The rewrites are exact, including their changed masses and domains.

Every requested profile remains a fixed positive expression with heads below the relevant limit L: only affine maps, fixed lower heads and h_ω are composed. Thresholds are profile dependent, and are computed by following these finitely many smaller children.

8.3 A common well-founded measure

Expand sharing when counting occurrences. Let h be the number of original source h_λ nodes across the two compared expressions, including their argument nodes. Let $\mathcal{Q}$ be the finite collection of principal indices in the Cantor decompositions of all original source heads and initial auxiliary kernels. Close it under finite successor-exponent chains, stopping at exponent one or a limit exponent. No rule introduces a principal outside this finite set. List it increasingly as $\pi_0 < \cdots < \pi_{m-1}$ and assign a kernel occurrence of index π_i weight ω^{m-1-i}. Let W be their natural sum across the pair. Use the lexicographic measure

$$(h, W). \tag{8.3}$$

Replacing an original tag by its peak removes that displayed h node, retaining its argument only once in each child comparison. In P/P and R/P children, other siblings are not duplicated. Thus h decreases. Equal-principal comparison removes both outer kernels, decreasing W. Mixed-limit comparison removes the lower outer kernel, also decreasing W. A combined mixed-successor step removes the opposite lowest-rank M occurrence and adds only finitely many occurrences of principals strictly greater than M, retaining the original higher occurrence internally. Their total weight is strictly smaller than one M weight. Hence W decreases. Affine or closed leaves stop.

Since h is natural and $W < \omega^m$, every branch terminates. Branching is finite, so the comparison tree is finite. The certificate uses that same tree: each invocation in its case analysis belongs to a strictly smaller comparison child or is a direct supplier application. Thus no unchanged comparison state calls its own certificate. Mutual induction on (8.3) proves both comparison correctness and (8.2), establishing Theorem 8.1.

8.4 Closure and exact finite exceptions

The two sorts are closed under every recursive child. Peak replacement introduces a finite kernel word on a positive argument plus its fixed norm constant. Successor expansion introduces only a longer kernel word. No signed inner offset, exposed count, division, inverse iterate or variable ordinal index occurs. Increasing a mass changes fixed domain constants without changing the finite rank set or measure.

Example 8.2 (A nested comparison with the counts retained)**.** Consider $E = h_{2\omega}(x)$ and $F = h_\omega(h_\omega(x))$. Deleting the two ω blocks gives

$$E = f(f(x+4)) + 2, \qquad F = f(f(x+2)+3) + 1.$$

Thus their peaks are respectively $f(f(x+4))$ and $f(f(x+2)+3)$. The equal-outer-ω rule compares the arguments $f(x+4)$ and $f(x+2)+3$. The latter positive nonlinear expression lies strictly between its peak $f(x+2)$ and the next range value on the computed tail. Comparing $f(x+4)$ with that peak reduces to the affine comparison $x+4 > x+2$. Hence $f(x+4) > f(x+2)+3$, and the original signed-row rule returns $E > F$ on the effective tail. The offset 3 has been paid by a range-cell test, rather than deleted as negligible without proof. The final procedure evaluates the finitely many earlier inputs exactly.

Example 8.3 (Why the mass key changes at degree three)**.** At the same genuine input, the masses 1 and 2 give equal ω^2 kernels because their ceilings after division by two agree. For a parent ω^3 comparison with mass difference one, the first common ω^2 block can likewise tie when its retained child masses are an odd a and $a+1$. The next block reduces both child masses by three; its ceiling classes are then distinct. The guaranteed two common blocks ensure that those reduced masses remain nonnegative. This is the two-block mechanism in Theorem 7.3, and explains why the algorithm uses neither strict mass order at degree two nor a ceiling equivalence at all higher degrees.

Corollary 8.4 (One-input quantified comparison)**.** *For any finite Boolean combination of equality, weak-order and strict-order atoms between expressions of $\mathcal{P}_1$, each of its existential and universal closures over $x \in \mathbb{N}_{>0}$ is decidable.*

Proof. Apply Theorem 8.1 to every atom and take the maximum tail bound. The Boolean truth value is constant beyond that bound. Evaluate the original positive input expressions exactly at every earlier positive integer. For an existential statement, a true prefix instance or a true tail suffices; for a universal statement require all prefix instances and the tail. Evaluate original expressions, rather than auxiliary kernels before their raised genuine domains. □

9 Consequences for canonical assignments and exact limits

The Actual-source forward translation theorem in the companion foundations paper [2] proves the following assertion for original undecorated Gödel-T terms. If all free term variables have ground type, every positive canonical coordinate is a closed computable ordinal, and coordinate zero is effectively translated, valuation preservingly, into $\mathcal{P}_d$. The source may have

arbitrary temporary higher-type binders, recursors, non-normal syntax and output type. The result is a generating invariant of the actual source translation; it is not an assertion that every ambient ordinal-expression grammar has closed positive coordinates. In particular, it does not include the decorated auxiliary R_t syntax of the extended rewrite system.

Corollary 9.1 (Ground-open source equality)**.** *Universal equality of literal canonical assignment vectors is decidable for any original source pair whose free names all have ground type.*

Proof. Construct their canonical assignments by the literal syntax rules. Compare the closed positive coordinates exactly and compare their zero-coordinate normal forms by Theorem 2.3. A positive mismatch refutes vector equality. Equal positive coordinates and equal normal forms prove it. If the normal forms differ, (2.2) supplies an admissible positive ground-coordinate countervaluation. This is coordinate substitution, not replacing a free variable by a source numeral and recomputing its assignment. □

For different vector levels, pad with zeros through their maximum level, as in the original assignment convention. If A, B have only one shared ground free name X, write their residuals as $E_A(x), E_B(x)$ and put

$$U_{\leq} = \bigwedge_{j\geq 1}([A]_j \leq [B]_j), \qquad U_{=} \bigwedge_{j\geq 1}([A]_j = [B]_j), \tag{9.1}$$

where only finitely many padded coordinates are involved. The original weak comparison is $U_{\leq}$ together with zero-coordinate weak comparison; the original strict comparison is $U_{\leq}$ together with *strict* zero-coordinate comparison. It is neither lexicographic order nor strictness at every coordinate.

Corollary 9.2 (Whole one-ground-name source comparison)**.** *Every finite Boolean combination of these pointwise vector equality, weak-comparison and strict-comparison atoms is decidable under one existential or universal quantifier over its shared positive ground coordinate. Arbitrary original internal types and output levels are allowed.*

Proof. Compute the closed guards (9.1) and put them inside their respective pointwise Boolean atoms. Apply Corollary 8.4 to the zero-coordinate expressions. Ground free variables have precisely the positive-natural admissible coordinate values of the original definitions [1, Definitions 2.5–2.8]; no extra witness variable or observer translation is needed. □

One free ground name in each term is insufficient if the names differ: the union of names across the entire query must lie in a single $\{X : o\}$. The one-variable support hypothesis is used explicitly in Lemma 3.1 and the range-cell rule.

For two independent inputs, a child peak depending on a small coordinate need not dominate an affine term depending on a different large coordinate. Our finite-zero and sign proofs therefore do not settle multivariate weak or strict comparison. The multivariate identity theorem likewise does not decide existence of a valuation solving a nonidentity equation; for example $x = y$ has solutions although its normal forms differ. Functional free variables and their admissibility constraints are outside the source corollaries. No auxiliary numerical kernel is claimed to be the direct assignment image of a source term.

The algorithms are mathematical decision procedures. Their finite ordinal descent and finite syntax recursion establish termination, but evaluating closed constants or a finite exceptional prefix can be far beyond practical computation. The proof requires no evaluation of a giant F_5 value. The role of the strong fixed control is structural: sparse peaks isolate positive sums, and packed ordinal reserves pay entire lower profiles.

References

[1] Gunnar Wilken and Andreas Weiermann. Derivation lengths classification of Gödel's T extending Howard's assignment. *Logical Methods in Computer Science* **8**(1:19) (2012), 1–44. doi:10.2168/LMCS-8(1:19)2012. Official article and full text. The 2012 revised article is the version used throughout.

[2] *The Wilken–Weiermann Comparison Problem: Foundations and a Guide to the Results.* Companion working manuscript, 2026. Included manuscript.

[3] *A Type-Confined Termination Ordering for Gödel's T.* Companion working manuscript, 2026. Included manuscript.